

REGINA RUIZ

BIOELECTRICAL ENGINEER— EMBEDDED SYSTEMS | MIXED-SIGNAL PCB DESIGN | LAB INSTRUMENTATION

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TECHNICAL SKILLS

Embedded Hardware & PCB: KiCad schematic capture & PCB layout, **mixed-signal circuit design**, ERC/DRC, Arduino Uno/Nano, ESP32, A4988 drivers, CNC shields, L298N drivers, NEMA 17 motors, microstepping, Vref calibration, **oscilloscope & multimeter** bench measurement, FSR sensors, I²C, LoRaWAN, BLE

Firmware & Programming: C/C++, Arduino IDE, AccelStepper/Stepper.h, Python, MATLAB, Linux CLI, Git/GitHub, HTML, Web Bluetooth/BLE

CAD & Prototyping: SolidWorks, 3D printing, rapid prototyping, electromechanical integration, design-for-assembly, hardware validation

Bioinformatics & Genomics: Oxford Nanopore analysis, MinKNOW, Dorado, SAMtools, Bismark, BEDTools, DNA methylation & epigenetic clocks, pipeline development

Wet Lab: DNA extraction, library prep, Nanopore sequencing, flow-cell operation, genotyping, hydrogel characterization, microscopy, microfluidics

Languages: Spanish (Native), English (Fluent)

EDUCATION

Lehigh University — B.S. Bioengineering, Bioelectronics Track

Bethlehem, PA | Expected May 2027

Relevant Coursework: Circuits, Signals & Systems, Physiology, Engineering Programming, Biomaterials, Statistics

ENGINEERING EXPERIENCE

Lab Engineering & Bioinformatics Intern | *Granatum Bioworks — Monterrey, México*

Jun 2026 – Present

- Cut lab equipment costs over 90% versus commercial alternatives by designing, fabricating, and validating two custom instruments (MegaRuptor DNA-shearing device, Hula Mixer), integrating SolidWorks CAD, 3D-printed housings, stepper-motor drive systems, and custom PCB electronics.
- Delivered a zero-violation, manufacturing-ready PCB for a mixed-signal embedded system by independently completing schematic capture, part selection, layout, and design verification in KiCad for the MegaRuptor's Arduino-shield board (A4988 stepper driver, 12V power stage, hardware fail-safe circuit).
- Reduced motor-driver stalls and failures by 85% by debugging and calibrating A4988/L298N stepper-driver current — using a bench multimeter and oscilloscope to measure Vref and verify switching behavior — and tuning microstepping configuration; documented the build via Git/GitHub with BOMs and decision logs.
- Built a Nanopore sequencing bioinformatics pipeline (MinKNOW, Dorado, SAMtools, Bismark, BEDTools) to process DNA methylation data and estimate biological age via epigenetic clocks, supporting the EpiSignal platform (submitted to FONLIN 2026) and Auragen's raspberry breeding program.

Bioelectronics Engineer, Wearable Capstone | *Good Shepherd Rehabilitation — Bethlehem, PA*

Jan 2026 – Present

- Enabled real-time quantification of weight-bearing asymmetry across four foot regions by designing an 8-channel FSR sensor array on Arduino Nano ESP32 hardware, sampling at 20 Hz with threshold-triggered feedback.
- Delivered a browser-based clinical dashboard for live pressure and asymmetry visualization by building a BLE embedded acquisition pipeline streaming real-time sensor data from device to a Web Bluetooth API clinician-facing interface.
- Defined firmware and validation requirements (sampling rate, force-calibration coefficients, threshold-triggered feedback) and contributed to wearable packaging decisions spanning insole placement, strain relief, and battery housing.

Researcher, CareAlert Team | *Global Social Impact Fellowship, Lehigh University — Bethlehem, PA*

Jan 2025 – Present

- Supported 100 paired devices across a 5–10 km range with sub-5-second message delivery by defining hardware architecture, reliability specs, and battery requirements for a low-cost, off-network maternal-health paging system for rural clinics in Sierra Leone.
- Built and evaluated breadboard prototypes, translating field constraints into reliability, battery, and alert specifications; work published at IEEE GHTC 2025 and presented at DMD2026.

ADDITIONAL RESEARCH & PRODUCT EXPERIENCE

Research Intern, Microfluidic Systems | *Laboratory of the Future — Bordeaux, France*

Jun – Aug 2025

- Characterized flow-rate and residence-time behavior of a pressure-controlled microfluidic system (~100–700 mbar) by building MATLAB analysis pipelines for microscopy data, supporting skin-on-chip device validation for an international research team.

Research Assistant, Materials Characterization | *Prof. Daniel Ou-Yang, Lehigh University — Bethlehem, PA*

Jan – Aug 2025

- Quantified hydrogel mechanical behavior across ionic and mechanical conditions by running microscopy, particle-tracking, and rheology experiments, then analyzing multivariable datasets in MATLAB and Python.

Co-Founder, Product Lead | *Alimenta — Monterrey, México & Bethlehem, PA*

Aug 2025 – Present

- Took a consumer health app from concept to shipped product on the Apple App Store, owning feature definition, UX/UI, and usability testing end to end.

PUBLICATIONS

O. Meyer, **R. Ruiz**, et al., "Designing an Off-Network Paging System to Improve Continuity of Patient Care in Low-Resource Settings," IEEE Global Humanitarian Technology Conference, 2025.

P. Natarajan, L. Hamamoto, O. Meyer, **R. Ruiz**, et al., "The Power of a Ping," Design of Medical Devices Conference (DMD2026), Minneapolis, MN, 2026.